

AEXIS DIAGNOSTICS - Technical Documentation

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Date: August 29, 2026

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1. Executive Summary

AEXIS DIAGNOSTICS is a clinical-grade AI prediction system engineered for high-stakes medical profiling. It leverages historical patient health records, vital signs, and blood panel data to predict the likelihood of disease onset, assisting medical professionals in proactive triage.

The system operates through a strict, medical-themed UI environment, ensuring data integrity and clear diagnostic readouts.

2. Clinical Data Processing

Medical data is notoriously noisy. The data pipeline was constructed with healthcare standards in mind.

- Outlier Detection: Filtered anomalous vitals (e.g., impossible systolic blood pressure readings) using Interquartile Range (IQR) gating.
- Imputation: Handled missing blood panel values using K-Nearest Neighbors (KNN) imputation, preserving the multi-dimensional clinical relationships.
- Feature Selection: Utilized Recursive Feature Elimination (RFE) to isolate the most critical biomarkers driving disease pathology.

3. Diagnostic Model Architecture

High-stakes medical predictions require models that are both highly accurate and highly interpretable.

- Algorithms: Ensembled Gradient Boosting Machines (XGBoost) and Random Forest Classifiers.
- Interpretability: Integrated SHAP (SHapley Additive exPlanations) values to extract feature importance, allowing doctors to understand **why** the AI made a specific diagnosis.
- Calibration: Applied Platt Scaling to convert model log-odds into true clinical probabilities (percentages).

4. Performance & Validation

The system was evaluated with a strong emphasis on Recall (Sensitivity) to minimize False Negatives (failing

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to diagnose a sick patient).

- Achieved top-tier F2-Scores (which heavily penalizes missed diagnoses).
- K-Fold Cross Validation ensured the model generalizes across diverse patient demographics without overfitting.

5. Frontend Clinical Interface

The AEXIS DIAGNOSTICS web application provides a futuristic, secure terminal for doctors.

- Patient Intake Forms: Allows manual entry of vitals and demographics, directly hooked into the prediction API.
- Clinical Reports Dashboard: Visualizes risk vectors and historical patient data.
- Unified Routing: The Flask backend securely serves multiple endpoints (/model-lab, /database-registry) with zero-latency template caching, optimized for immediate clinical use.